

Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 5

TITLE: Designing Visually Pleasing and Effective Screen-Based Controls: An Exploration of Color, Typography, and Layout Principles.

Objective:

Understand various types of wireframes and develop a wireframe for sample application.

Expected OUTCOME of Experiment: CO 4

Analyze tools and techniques for prototyping and designing applications

Books/ Journals/ Websites referred:

<https://www.figma.com/resource-library/color-combinations/>

<https://www.canva.com/learn/100-color-combinations/>

Background Theory:

Visual design is the aesthetic aspect of user interface design, focusing on how a product looks and feels. It plays a crucial role in user perception, brand identity, and overall user experience. Beyond mere aesthetics, good visual design guides the user's eye, communicates hierarchy, and evokes appropriate emotions.

Key elements and principles of visual design include

1. Layout and Composition:

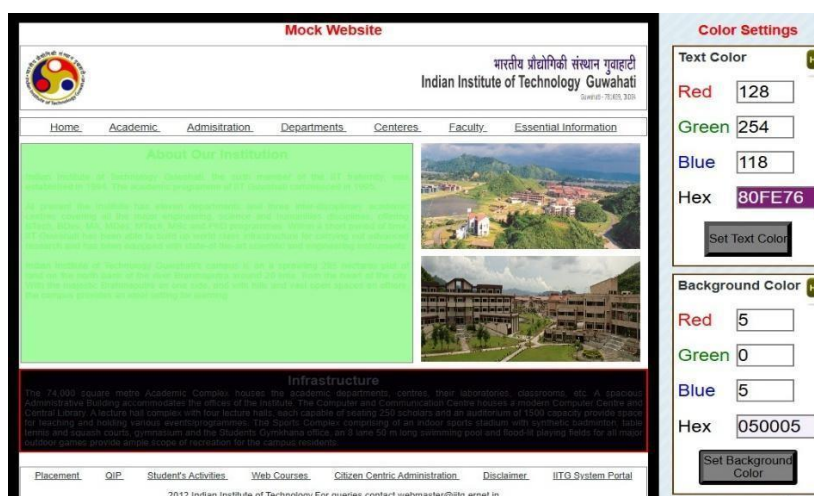
- o **Balance:** Distributing visual weight evenly to create stability. Can be symmetrical or asymmetrical.
- o **Alignment:** Arranging elements so their edges or centers line up, creating order and professionalism.
- o **Proximity:** Grouping related items together to create visual units and reduce clutter.
- o **Repetition:** Consistent use of visual elements (e.g., colors, fonts, shapes) to create unity and reinforce branding.

- o **Contrast:** Using differences in size, color, shape, or texture to make elements stand out and create visual interest.
 - o **Visual Hierarchy:** Arranging elements to show their order of importance, guiding the user's attention.
2. **Color Theory:**
- o **Color Wheel:** Understanding primary, secondary, and tertiary colors, and relationships like complementary, analogous, and triadic.

- o **Color Psychology:** The emotional and psychological effects of different colors (e.g., blue for trust, red for urgency).
 - o **Color Palettes:** Creating harmonious sets of colors for a product, often including a primary, secondary, accent, and neutral colors.
 - o **Accessibility:** Ensuring sufficient color contrast for readability, especially for users with visual impairments (WCAG guidelines recommend a contrast ratio of at least 4.5:1 for normal text).
3. **Typography:**
- o **Font Selection:** Choosing appropriate typefaces (serif, sans-serif, display, monospace) that match the product's personality and purpose.
 - o **Readability vs. Legibility:** Readability refers to the ease of reading long blocks of text; legibility refers to the ease of distinguishing individual characters.
 - o **Hierarchy:** Using different font sizes, weights, and styles to indicate importance and structure.
 - o **Line Height (Leading), Letter Spacing (Kerning/Tracking):** Adjusting spacing for optimal readability.
4. **Screen-Based Control Design:**
- o **Buttons:** Visual cues for clickability (e.g., shadows, gradients, fills, hover states). Clear labels and appropriate sizing.
 - o **Input Fields:** Clear labels, placeholder text, visual feedback for focus, error states, and success states.
 - o **Sliders, Toggles, Checkboxes, Radio Buttons:** Intuitive visual representation of their state and functionality.
 - o **Navigation Elements:** Clear visual distinction for active states, hover states, and disabled states.

Laboratory Work:

BAD UI



Mock Website



भारतीय प्रौद्योगिकी संस्थान गुवाहाटी
Indian Institute of Technology Guwahati
Guwahati - 78103, India

Home Academic Administration Departments Centres Faculty Essential Information

About Our Institution

Indian Institute of Technology Guwahati, the sixth member of the IIT fraternity, was established in 1994. The academic programme of IIT Guwahati commenced in 1995.

At present the Institute has eleven departments and three inter-disciplinary academic centres covering all the major engineering, science and humanities disciplines, offering BTech, BDes, MA, MDes, MTech, MSc and PhD programmes. Within a short period of time, IIT Guwahati has been able to build up world class infrastructure for carrying out advanced research and has been equipped with state-of-the-art scientific and engineering instruments.

Indian Institute of Technology Guwahati's campus is on a sprawling 285 hectares plot of land on the north bank of the river Brahmaputra around 20 kms. from the heart of the city. With the majestic Brahmaputra on one side, and with hills and vast open spaces on others, the campus provides an ideal setting for learning.

Infrastructure

The 74,000 square metre Academic Complex houses the academic departments, centres, their laboratories, classrooms, etc. A spacious Administrative Building accommodates the offices of the Institute. The Computer and Communication Centre houses a modern Computer Centre and Central Library. A lecture hall complex with four lecture halls, each capable of seating 250 scholars and an auditorium of 1500 capacity provide space for teaching and holding various events/programmes. The Sports Complex comprising of an indoor sports stadium with synthetic badminton, table tennis and squash courts, gymnasium and the Students Gymkhana office, an 8 lane 50 m long swimming pool and flood-lit playing fields for all major outdoor games provide ample scope of recreation for the campus residents.

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2012 Indian Institute of Technology For queries contact webmaster@iitg.ernet.in

Color Settings

Text Color HELP

Red

Green

Blue

Hex

Background Color HELP

Red

Green

Blue

Hex

Mock Website



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GOOD UI :

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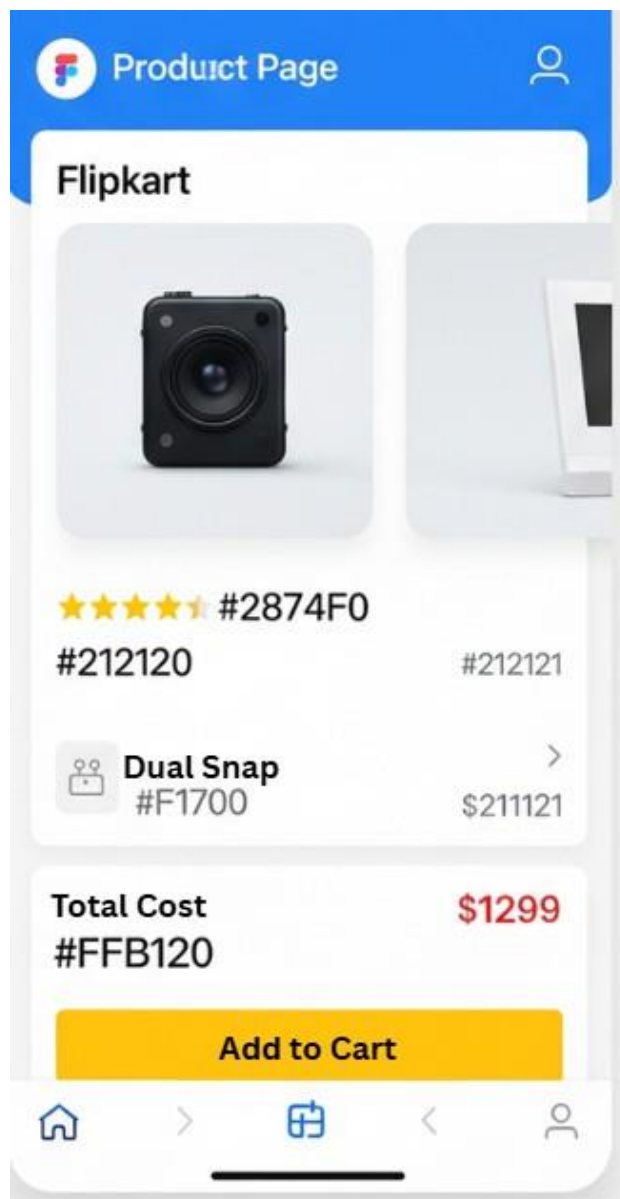
Activity:

1. Choose one of the following scenarios for your wireframing project. You will design the main screens for this application.
 - Scenario A: Simple E-commerce Product Page: Design a product detail page for an online store, including product image, name, price, description, "add to cart" button, quantity selector, and related products section.
 - Scenario B: Basic To-Do List Application (Web or Mobile): Design the main screen for a To-Do list app, allowing users to add tasks, mark them as complete, and view existing tasks. Consider basic navigation.

- Scenario C: User Profile Page for a Social Media App: Design a user's profile page, including profile picture, name, bio, number of followers/following, and tabs for different content types (e.g., posts, photos).

2. Justify your color choices based on color psychology and the desired brand mood.

Flipkart Wireframe Activity



Primary Colors (Flipkart Brand)

- **#2874F0 (Flipkart Blue)** → trust, professionalism, tech-driven.
- **#FFB700 (Flipkart Yellow)** → optimism, draws attention to CTAs like “Add to Cart”.
- **#FFFFFF (White/Neutral)** → clean background, ensures readability.
- **#212121 (Dark Grey)** → for text, modern and minimal contrast.

Mood/Brand Fit:

This palette reflects Flipkart’s actual brand identity: **reliable, energetic, and approachable**. Blue = trust & tech, Yellow = excitement, White = simplicity.

Accessibility Check:

- Blue text (#2874F0) on white → WCAG AA passes.
- Yellow on white → needs dark borders or bold text for readability.

Screens to Design

1. **Home Screen** → Search bar, categories, banners, trending products.
2. **Product Page** → Image, name, price, description, Add to Cart, Buy Now.
3. **Cart Page** → Selected items, total cost, checkout button.

Typography

- **Headings:** Bold Sans-serif (e.g., Poppins / Roboto)
- **Body Text:** Regular Sans-serif, 14–16px, dark grey.
- **CTA Buttons:** Medium weight, uppercase.

-
3. Create visually distinct states for interactive elements (e.g., normal, hover, active/pressed, disabled states for buttons).

Buttons:

- Normal → Solid Blue (#2874F0), white text.

- Hover → Darker Blue.
- Active → Pressed effect (inset shadow).
- Disabled → Light grey, low opacity.

- **Input Fields (Search, Login):**

- Normal → White with grey border.
- Focus → Blue border glow.
- Error → Red border + helper text.

Other Controls:

Toggles/sliders should follow the same palette for consistency.

4. Ensure input fields have clear labels and visual cues for focus.
5. Design other controls (toggles, sliders, etc.) to be intuitive and visually appealing
6. **Document each interaction tabulate meticulously:**

Documentation Table (Flipkart)

<u>Design Decision</u>	<u>Visual Design Principles</u>	<u>Feedback Category</u>	<u>Challenges / Improvement</u>
Primary Color Choice (Blue + Yellow)	Contrast, brand identity	Accessibility & branding	Yellow text contrast needs bold weights
Typography (Poppins/Roboto)	Hierarchy, legibility	Readability	Some small text may need scaling for mobile
Button Style	Consistency, affordance	Interactivity	Add clearer hover animations
Navigation (Bottom bar + Search on top)	Balance, proximity	Usability	Icons must have labels for clarity
Product Page Layout	Alignment, hierarchy	Scan-ability	Prevent info overload by grouping sections
Help/Support UI	Visibility, accessibility	Support	Should include floating chat/help button

7. Identify common themes or recurring types of usability issues across different products.
 - **Poor contrast:** text hard to read in sunlight.
 - **Hidden help features:** users struggle to find assistance.
 - **Unclear button states:** uncertainty about whether an action was triggered.
 - **Overloaded screens:** too much info reduces focus.
8. Reflect on how these issues could have been prevented during the design and engineering phases.
 - Early **WCAG contrast testing**.
 - **User journey mapping** to prioritize help visibility.
 - **Prototyping with micro-interactions** (hover, pressed states) before dev handoff.
 - Frequent **usability testing** with real users in varied conditions (low light, glare, etc.).

Post Laboratory Questions:

Following questions are to be handwritten and attach photographs of the answer in this document.

1. Identify the primary, accent, and background colors used in your screen design.
2. Explain how these color choices align with the mood, purpose, or branding of your application.
3. Select a common web control (e.g., button, input field, checkbox).
4. Describe its appearance in the default (normal) state.
5. Explain how its appearance changes during user interactions such as hover, click (active), and disabled states.
6. Describe any visual elements you simplified or removed to adapt your design for different screen sizes (e.g., mobile, tablet, desktop).
7. How did you maintain both usability and aesthetic consistency across various devices?

<u>Ans. 1</u>	Primary - blue → Trust, reliability Accent - Green → Success, confirmation action Background - light gray / white → clean, minimal look.
<u>Ans. 2</u>	① Blue builds trust, E.g. Ticket Booking ② Green highlight positive actions. E.g. Payment success. ③ Neutral background keeps focus on main content.
<u>Ans. 3</u>	Default state → Solid deep blue background, white text, rounded corners, subtle shadows.
<u>Ans. 4</u>	Hover state → Slightly darker blue background, cursor changes to pointer.
<u>Ans. 5</u>	Active state → Pressed effect with a darker stroke and inset shadow. Disabled state → Light gray background, faded text.
<u>Ans. 6</u>	Mobile → Collapsed navigation in hamburger menu. Tablet → Reduced grid columns, larger than touch targets. Desktop → Full navigation bar with labels.
<u>Ans. 7</u>	1) Consistent color palette, fonts and spacing. 2) Clear information, Header → Content → Action. 3) Responsive layout ensure readability across all screens.